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Development of an autonomous system based on computer vision and open software-hardware platforms, applied to an educational game for inclusive basic education

Carlos Calderon-Cordova
Dept. of Computer Science and Electronics
Universidad Tecnica Particular de Loja
Loja, Ecuador
cacalderon@utpl.edu.ec
ORCID: 0000-0002-7071-7088

Santiago Sarmiento-Sinche
Dept. of Computer Science and Electronics
Universidad Tecnica Particular de Loja
Loja, Ecuador
sjsarmiento@utpl.edu.ec

Roger Sarango
Dept. of Computer Science and Electronics
Universidad Tecnica Particular de Loja
Loja, Ecuador
rasarango1@utpl.edu.ec
ORCID: 0000-0003-0750-1169

***Abstract*—Taking into account that technology must have a great impact on educational inclusion, this work addresses the development of a computer vision based system for visually impaired children as an alternative to the teaching and learning of mathematics through a board game, the low-cost technological system monitors compliance with the activities and gives feedback to the user. The hardware architecture of the implemented prototype consists of the following elements: Raspberry Pi3 model B electronic card, a low cost camera, a fixed boardgame, tabs with a numerical value and symbols that fit into each square of the boardgame, a keyboard by which the user sends instructions to the device, a speaker and display. The set of algorithms developed in Python and OpenCV include: ROI extraction, shadow elimination, conversion to grayscale, blurring, binarization, erosion, equalization, segmentation, and optical character recognition. For the validation of the prototype in the identification of the game tabs based on symbols and characters consisted of 12 tests for each character, resulting in a total of 168 tests, all tests were performed under illuminance conditions of 16 lux and 103 lux. As a result, for 16 lux brightness conditions the percentage of sensitivity is 80.35%, accuracy is 83.35% and overall accuracy is 96.55%. While, for 103 lux conditions the sensitivity percentage is 97.61%, accuracy is 99.40% and precision is 98.20%. Finally, the prototype was tested on visually impaired children and the user experience was satisfactory.**

Keywords—computer vision, Optical Character Recognition, OpenCV, image processing, inclusive education.

I. INTRODUCTION

According to the National Institute of Statistics and Censuses (INEC) [1]; in the Ecuador there are about 186,000 people with visual impairment. Of these17,000 are children with visual impairment of school age (under 12 years old). But only 77% have access to school education [2]. Special education institutions lack material for children with visual impairment and special teaching staff. Through Article 47 of the Organic Law of Intercultural Education [2], the Ecuadorian government states that the inclusion and integration of people with disabilities in educational institutions will be guaranteed, eliminating barriers to learning. To complement this, Articles 229 and 230 states that specialized professionals must define the most appropriate modality for each student and adapt the learning standards according to the needs of each student.

In the context of teaching and cognitive education, interactive games are an indispensable tool for learning from an early age, thus stimulating creativity and logical reasoning. The purpose of this article is to develop and implement a prototype to automate the teaching-learning process in students with visual impairment, where the device guides, supervises, and provides feedback on the activities developed by the student. The prototype integrates elements of computer vision and software engineering.

This paper addresses three main stages for the construction of the proposed objective: section II summarizes the related work, section III presents the design and development of algorithms and the software-hardware architecture of the prototype, section IV addresses the implementation and testing in different lighting environments.

II. RELATED WORKS

In the technological market and in academia, innovative platforms applied in the educational field have been developed, which serve as an example for the present device. The following is a brief summary of related works and devices, as well as the approach of the tool created.

In [3] the game and application developer developed educational games designed for children ages 5 to 11 years old, based on which it encourages learning in key areas such as creative problem solving, art, science, technology, engineering and mathematics.

In [4] computer interfaces for board games are generally based on the visual channel. The paper discusses a multipurpose

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system that especially allows blind and deafblind people to play chess or other board games over a network, thus reducing their barrier of due to disability. The proposed device is capable of giving a non-sequential representation of information; it also embeds a Braille display and an optical sensor along with a tactile interface system. The input is acquired by detecting finger pressure with a 64-electrode grid, while the output uses low-voltage electric current as stimulus.

In [5] allows visually impaired people to access graphical information on the computer screen through tactile perception. It applies the "braille box" in mathematics class focused on linear graphs to visually impaired children. The result shows that users can perform tasks such as: determining the slope and the intersection between two lines. The "braille box" aid system, consisting of a light pen and a tablet, is interfaced with a computer through specific TACTOS software, developed by the Perceptive Supplementation group. It can help visually impaired people to analyze linear graphics and allows the user to explore the shape of the object and receive tactile perception in the same hand.

In [6] an interactive game system for visually impaired people is presented, which combines haptics, audio and computer vision methods. It is a musical game where graphical interfaces such as OpenCV, OpenGL, and auditory methods are used to guide the visually impaired person in the game.

In [7], the design of a mathematical problem solving robot is presented as an educational robotics application. The robot scans and recognizes a math problem printed on a paper and then writes an answer on it, solves logical-mathematical problems and performs simple calculations. The recognition is done by means of computer vision algorithms to identify alphanumeric characters.

In [8] a mathematical game suitable for children and young people with visual impairment at primary and secondary school level was developed. The developed system has no need for graphics, the interaction features are complemented with audio and the game uses 2D spatial audio. The system was tested in two schools with blind and low-vision children obtaining great acceptance.

The educational game prototype of the present work is based on the initial work of [9], [10] and [11], which elaborates a proposal of didactic material that is adapted to children with visual impairment at the early school level (see Fig. 1), and that in turn can be used by a wide range of target audiences. In a later stage of development of the prototype, the authors investigated the use of games and toys as a pedagogical strategy to promote the psychomotor, logical-mathematical, and sensory development of children. In the Fig. 1, boxes 1 and 2 show the educational board game and the set of tabs; box 3 shows the commercial camera; box 4 shows the computer used for performing the procedure of machine vision; box 5 shows the front panel of the software.

The objectives of the game are focused on the child can: Become familiar with the Braille code. Recognize numbers up to 4 digits. Perform basic operations (+, -, x). Improve their mental and psychomotor development. Involve multiple senses in their learning process.

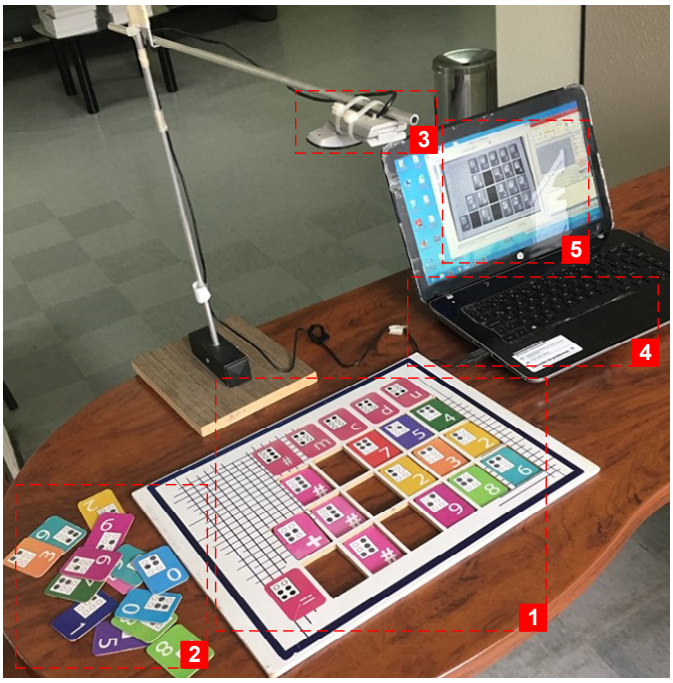


Fig. 1. Experimental setup of the initial prototype machine vision system [9].

III. Design and Development of Algorithms

Fig. 2 shows the software architecture containing the main functions developed with the respective programming environments and libraries. The tools used were Python 3.5, OpenCV 3.4.3, Numpy 1.12, and Pyttsx3 2.71.

The operations carried out are as follows:

A. *Image acquisition*

In this first stage the video input is read and converted into a matrix, so that a wide range of image formats can be handled. The video device identifier is placed, then the image is resized to fit the user interface.

B. *Region of interest (ROI) extraction*

The board is divided into 18 cells, distributed in 6 columns by 3 rows and whose dimensions are given by α and β, which represents the width and height of each cell. In addition, the boundaries of the board, defined by X_0, X_6, Y_0 and Y_3, are automatically determined. Once these three regions of interest have been obtained, the filtering and image processing phases can proceed.

C. *Shadow elimination*

Due to conditions of intensity and direction of the light source, as a consequence the board and the chips can generate shadows or regions of darkness in the image to be processed, this can cause problems in later stages of object and character recognition. To solve this problem, first, the image is dilated, then a Gaussian defocusing is applied with a 3x3 kernel to reduce the image noise, then the difference between the original image and the modified image is determined, and finally, the resulting image is normalized (see Fig. 3).

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D. *Conversion to grayscale*

The next step is to convert the resulting image to grayscale, OpenCV has a function that transforms an RGB image to grayscale (see Fig. 4a).

E. *Blurring*

The medium blurring stage is performed by applying a 5x5 kernel, thus enhancing the characters of the game pieces and eliminating the noise contained in the image (see Fig. 4b).

F. *Thresholding and Binarization*

The previously filtered image goes through the binarization stage, in which the Otsu threshold is applied. This reduces the grayscale to only two values (see Fig. 4c).

G. *Erosion and Equalization*

At this stage, the image is eroded, so that the characters of the chips have a finer and sharper texture, since the edges around them are reduced (see Fig. 4d). In the equalization stage, the resulting image is that of a sharper, black and white image, thus resembling a printed sheet of paper (see Fig. 4e).

H. *Segmentation*

Segmentation is performed by detecting contours and setting a minimum object area. This stage allows to separate each character and then apply the recognition individually.

I. *Optical character recognition*

Once the filtering and processing phase is finished, the image is suitable for OCR, using the Tesseract library. The OCR output is displayed in text format on the console for the rest of the functions that operate on the numbers entered by the user. The OCR engine used is based on Neural Networks.

J. *Voice Synthesis*

Once the character recognition is done, the next and last stage is the voice synthesis, for which the Pyttsx library is used. It replicates through audio the state of the game, the numbers entered, the selected operator, and the verification of the result.

IV. Prototype Implementation

The hardware architecture of the implemented prototype consists of the following elements: a fixed boardgame on which the state of the game will be evaluated, tabs with a numerical value and symbols that fit into each square of the board, a support to which the camera is attached, a keyboard by which the user sends instructions to the device, a speaker and display (see Figs. 5 and 6). The display is only for validation purposes of the prototype, the user of the game interacts only through the keyboard and receives feedback through audio.

The devices that make up the prototype are the following: Raspberry Pi3 model B electronic card, which is compatible with the implemented algorithms; KDC 600 (Viola HD) webcam, with a video resolution of 1600x1200 pixels; these devices are complemented with a 400 x 300 mm game board and 41 x 56 mm tabs.

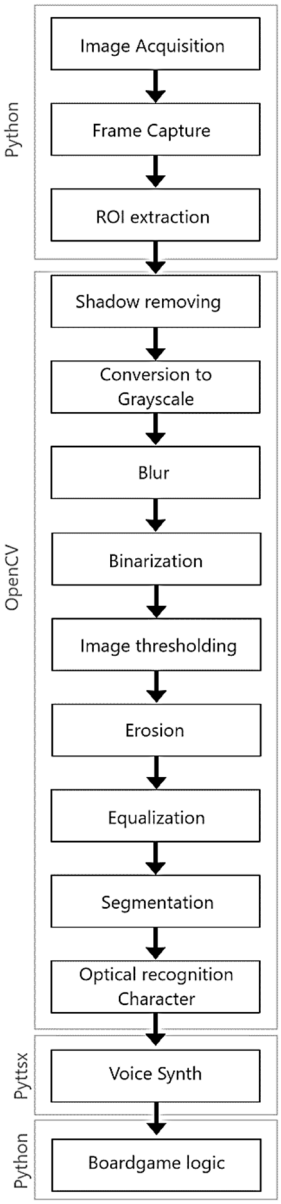


Fig. 2. Software architecture containing the main functions developed.

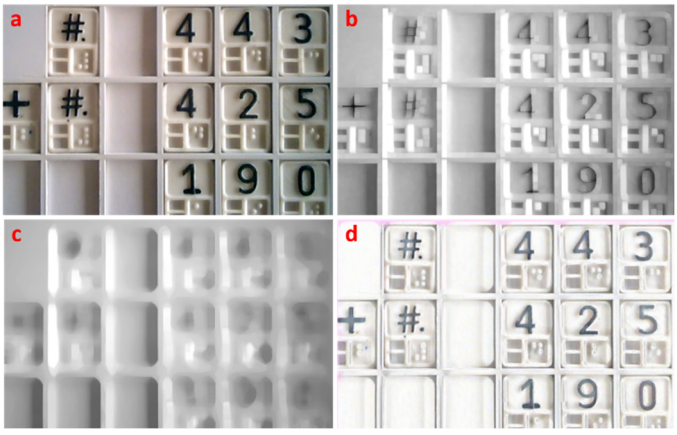


Fig. 3. Shadow elimination procedure

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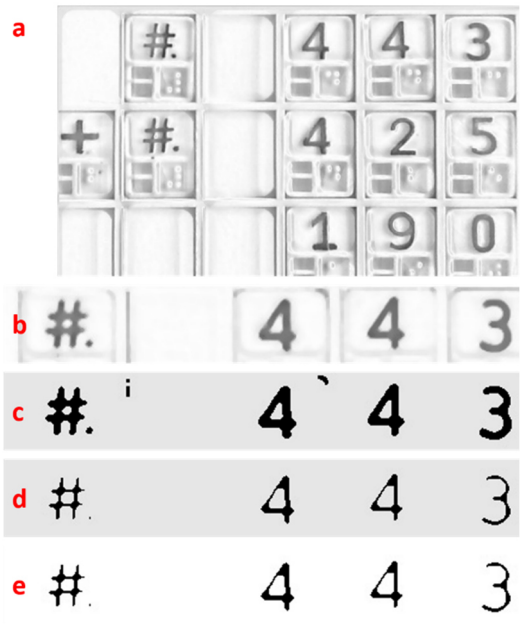


Fig. 4. Image operations: a) Grayscale image, b) Blurring, c) Binarization, d) Erosion, and e) Equalization.

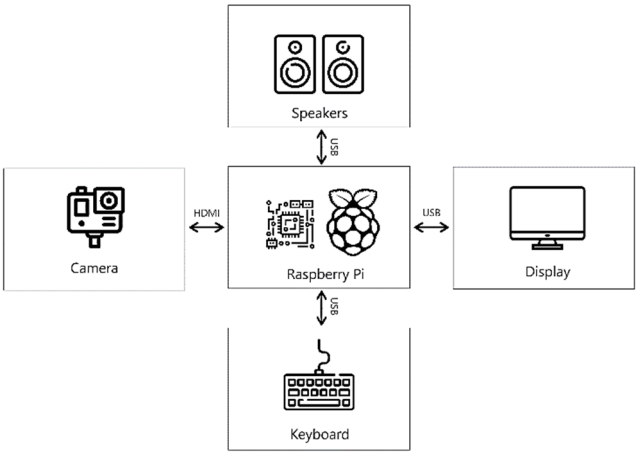


Fig. 5. Hardware architecture of the prototype



Fig. 6. Experimental setup of the implemented prototype.

V. Performance Evaluation

A. *Tests in different luminescent environments*

The tests are performed in different controlled lighting environments, in order to validate the effectiveness and functionality of the prototype during the execution of the game. The lighting levels in which the tests were performed were recorded using the mobile application "Lux Light Meter" available in Google PlayStore, which works using the light sensor of a smartphone. Figs. 7, 8, 9 and 10 shows the results of the evaluation of the prototype under illuminance conditions of 26, 37, 94, and 211 lux, respectively.

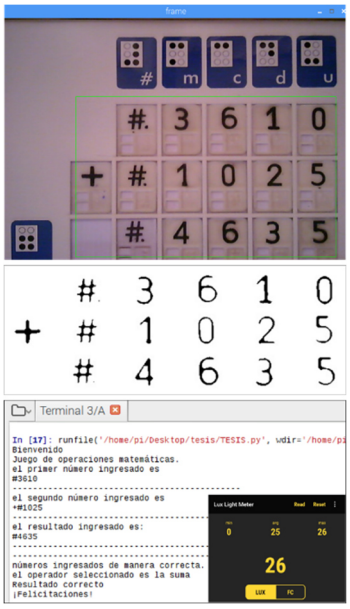


Fig. 7. Prototype performance test, a) image captured by the camera, b) result of the image processing algorithms, c) console result of the characters and symbols identified, d) illuminance measurement at 26 lux.

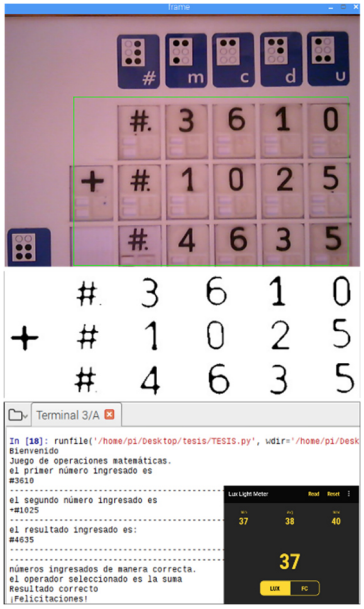


Fig. 8. Prototype performance test, a) image captured by the camera, b) result of the image processing algorithms, c) console result of the characters and symbols identified, d) illuminance measurement at 37 lux.

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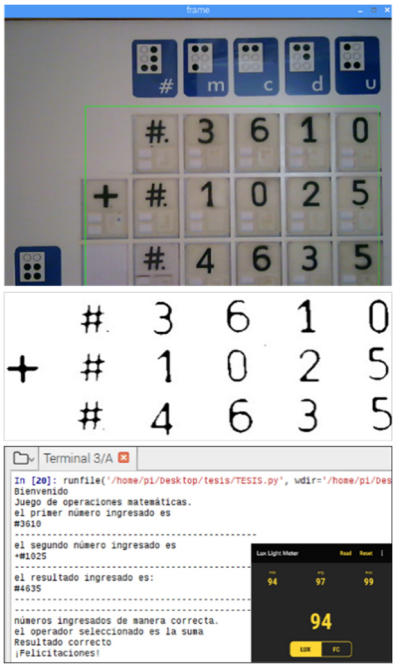


Fig. 9. Prototype performance test, a) image captured by the camera, b) result of the image processing algorithms, c) console result of the characters and symbols identified, d) illuminance measurement at 94 lux.

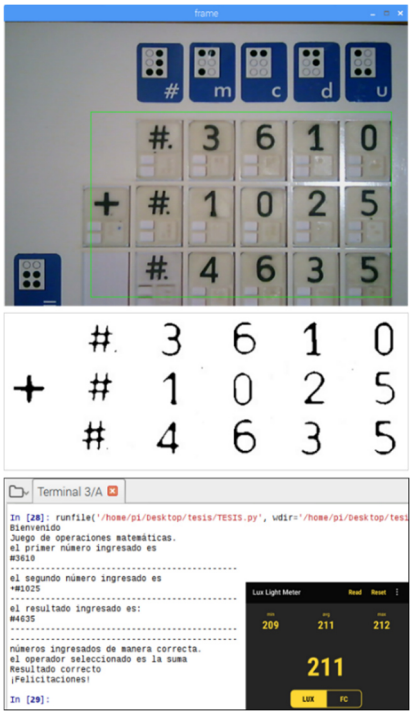


Fig. 10. Prototype performance test, a) image captured by the camera, b) result of the image processing algorithms, c) console result of the characters and symbols identified, d) illuminance measurement at 211 lux.

As a conclusion of the first tests, it was determined that the machine vision-based prototype has a correct hardware and software operation as long as the illuminance is higher than 26 lux. If the illuminance is lower than this value, the identification of characters and symbols presents incorrect results.

B. Character detection testing

The experiment performed for character detection consisted of 12 tests for each character, resulting in a total of 168 tests, all tests were performed under illuminance conditions of 16 lux and 103 lux. A value below the threshold and a value above the optimal operating threshold were chosen. Fig. 11 shows the percentage results for number and symbol recognition. For greater detail on the tests performed, Tables I and II show the confusion matrices, extracted under illuminance conditions of 16 and 103 lux.

As a result, for 16 lux brightness conditions the percentage of sensitivity is 80.35%, accuracy is 83.35% and overall accuracy is 96.55%. While, for 103 lux conditions the sensitivity percentage is 97.61%, accuracy is 99.40%, and precision is 98.20%.

Finally, the prototype was tested in a basic education institution for severe visually impaired children, this institution is located in Loja-Ecuador. The evaluations of the interaction between the child and the educational game board were satisfactory (see Fig. 12). Up to the moment of the initial tests, a formal evaluation instruction has not been defined, however the first uses by the children and the teachers project relevant results in the teaching and learning process.

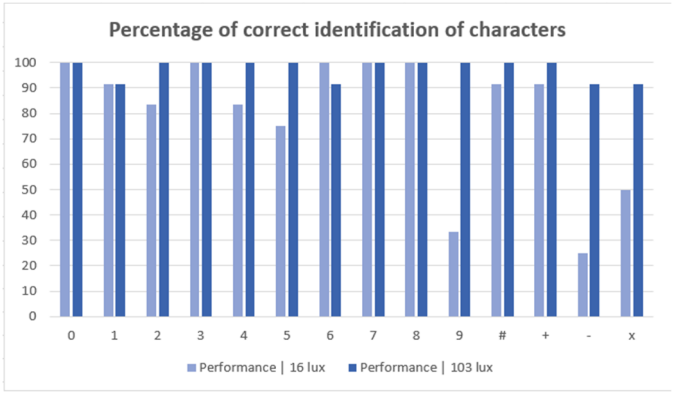


Fig. 11. Percentage results for Number and Symbol recognition.

TABLE I. CONFUSION MATRIX FOR CHARACTER DETECTION TESTING UNDER ILLUMINANCE CONDITIONS OF 16 LUX

	0	1	2	3	4	5	6	7	8	9	#	+	-	x
0	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
1	0 0%	11 91.66%	0 0%	0 0%	1 8.33%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
2	0 0%	0 0%	10 83.33%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
3	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
4	0 0%	0 0%	0 0%	0 0%	10 83.33%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
5	0 0%	0 0%	0 0%	0 0%	0 0%	9 75%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
6	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
7	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
8	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%
9	0 0%	0 0%	0 0%	1 8.33%	0 0%	0 0%	0 0%	0 0%	1 8.33%	4 33.33%	0 0%	0 0%	0 0%	0 0%
#	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	11 91.66%	0 0%	1 8.33%	0 0%
+	0 0%	0 0%	1 8.33%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	11 91.66%	0 0%	0 0%
-	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	3 25%	0 0%
x	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	6 50%

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TABLE II. CONFUSION MATRIX FOR CHARACTER DETECTION TESTING UNDER ILLUMINANCE CONDITIONS OF 103 LUX

	0	1	2	3	4	5	6	7	8	9	#	+	-	x
0	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
1	0 0%	11 91.66%	0 0%	0 0%	1 8.33%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
2	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
3	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
4	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
5	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
6	0 0%	0 0%	0 0%	0 0%	0 0%	1 8.33%	11 91.66%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
7	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%
8	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%	0 0%
9	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%	0 0%
#	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%	0 0%
+	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	12 100%	0 0%	0 0%
-	0 0%	1 8.33%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	11 91.66%	0 0%
x	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	11 91.66%



Fig. 12. First interaction tests between the prototype and children with severe visual impairment.

VI. CONCLUSIONS

A low-cost and portable system was designed and developed to be applied to teaching-learning processes in blind and visually impaired children. The technological prototype is based on a computer-based vision architecture, and makes use of open hardware and software platforms. This would guarantee replication anywhere and thus multiply the impact.

As a conclusion of the performance tests in different illuminance conditions, it was determined that the machine vision-based prototype has a correct hardware and software operation as long as the illuminance is higher than 26 lux. If the illuminance is lower than this value, the identification of characters and symbols presents incorrect results.

On the other hand, as a conclusion of the identification tests of the numbers and symbols of the educational game tabs, it was obtained that for 16 lux brightness conditions the percentage of sensitivity is 80.35%, accuracy is 83.35% and overall accuracy is 96.55%. While, for 103 lux conditions the sensitivity percentage is 97.61%, accuracy is 99.40% and precision is 98.20%. These values show the robustness and adaptability of the algorithm to varying illuminance conditions of the environment.

REFERENCES

[1] INEC (Instituto Nacional de Estadísticas y Censos), "Población por condición de discapacidad, según provincia, cantón, parroquia y área de empadronamiento," 2018. http://www.ecuadorencifras.gob.ec

[2] Ministerio de Educación del Ecuador, "Reglamento general a la ley orgánica de Educación Intercultural.," Reglamento general a la ley orgánica de Educación Intercultural., 2017. https://educacion.gob.ec/wp-content/uploads/downloads/2017/02/Reglamento-General-a-la-Ley-OrgAnica-de-Educacion-Intercultural.pdf

[3] OSMO, "Osmo interactive games," 2018. https://www.playosmo.com/es/faq/

[4] N. Caporusso, L. Mkrtchyan, and L. Badia, "A multimodal interface device for online board games designed for sight-impaired people," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 248–254, 2010, doi: 10.1109/TITB.2009.2034739.

[5] N. Sribunruangrit, C. Marque, C. Lenay, and O. Gapenne, "Graphic-user-interface system for people with severely impaired vision in mathematics class," *Conf Proc IEEE Eng Med Biol Soc*, vol. 7, pp. 5145–5148, 2004, doi: 10.1109/IEMBS.2004.1404432.

[6] M. Gaudina, V. Zappi, A. Brogni, and D. G. Caldwell, "Haptic, audio, and visual: Multimodal distribution for interactive games," *IEEE Trans Instrum Meas*, vol. 61, no. 11, pp. 3103–3111, 2012, doi: 10.1109/TIM.2012.2202071.

[7] Y. Park *et al.*, "Design and implementation of math-solving robot: Rank analysis for solving magic square puzzle," *International Conference on Control, Automation and Systems*, vol. 0, no. Iccas, pp. 1567–1571, 2016, doi: 10.1109/ICCAS.2016.7832510.

[8] F. Ferreira and S. Cavaco, "Mathematics for all: A game-based learning environment for visually impaired students," *Proceedings - Frontiers in Education Conference, FIE*, vol. 2015-Febru, no. February, 2015, doi: 10.1109/FIE.2014.7044493.

[9] C. A. Calderon, M. Guajala, J. Lanchi, L. Barba-Guaman, C. Bermeo and F. Rivas-Echeverría, "A machine vision system applied to the teaching of mathematics for blind or visually impaired children," *2018 IEEE International Conference on Automation/XXIII Congress of the Chilean Association of Automatic Control (ICA-ACCA)*, Concepcion, Chile, 2018, pp. 1-7.

[10] L. Barba-Guamán, P. A. Quezada-Sarmiento, C. Calderon-Cordova, L. Enciso and D. Guamán, "Development of software prototypes through model game jam and basic vision artificial techniques," *2017 12th Iberian Conference on Information Systems and Technologies (CISTI)*, Lisbon, Portugal, 2017, pp. 1-6

[11] C. Calderon Cordova, M. Quiñones Cuenca, B. Zuñiga Torres and E. Pineda Puglla, "System for interacting with board games for the blind". Patent Application WO2018173015.

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